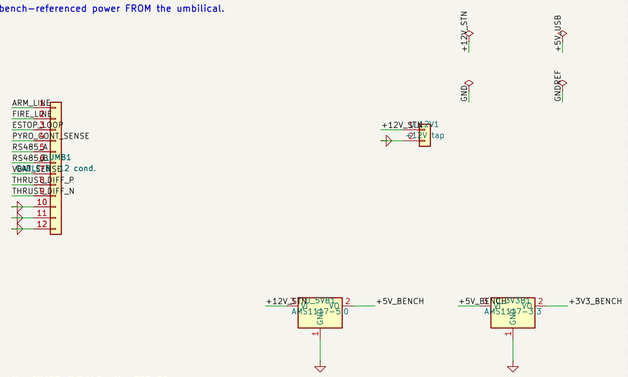
## 1. UMBILICAL — cable B to the bench

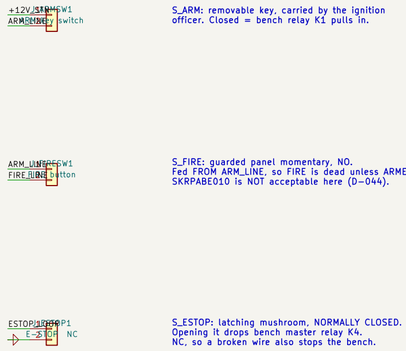
The station draws its bench-referenced power FROM the umbilical.




Bench-side rails: 5 V for the 6N137, 3.3 V for the SP3485.

## 2. CONTROLS — mechanical, panel-mounted

All four are PANEL controls wired to these terminals, not PCB parts.




S_ARM: removable key, carried by the ignition officer. Closed = bench relay K1 pulls in.

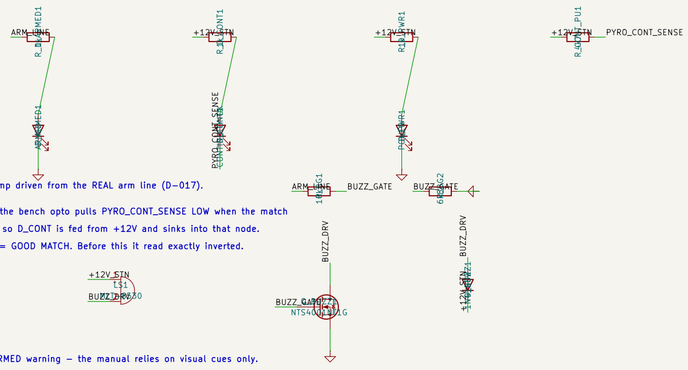
S_FIRE: guarded panel momentary, NO.
Fed FROM ARM_LINE, so FIRE is dead unless ARMED.
SKRPABE010 is NOT acceptable here (D-044).

S_ESTOP: latching mushroom, NORMALLY CLOSED.
Opening it drops bench master relay K4.
NC, so a broken wire also stops the bench.

H-2 FIX: the CONTINUITY test button is REMOVED.
As drawn it shorted PYRO_CONT_SENSE to ground, which is the lamp-good state, so the test reported a good match whether or not one was connected. A test that cannot fail is worse than no test.
Sensing is continuous and needs no button to initiate it.

OPEN DESIGN QUESTION: because sensing is continuous, current flows through the e-match whenever the bench is powered. Gating that needs a decision — see the audit, H-2.

## 3. INDICATORS & AUDIO




ARMED lamp driven from the REAL arm line (D-017).

H-1 FIX: the bench opto pulls PYRO_CONT_SENSE LOW when the match conducts, so D_CONT is fed from +12V and sinks into that node.
LAMP LIT = GOOD MATCH. Before this it read exactly inverted.

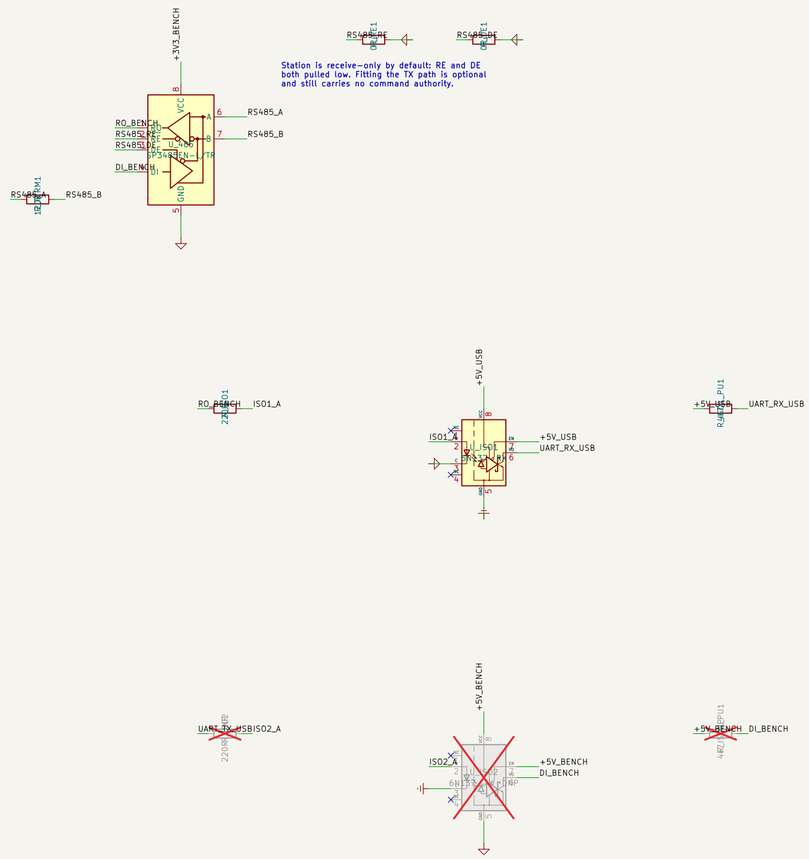
Audible ARMED warning — the manual relies on visual cues only.

## 4. TELEMETRY — RS-485 in, isolated, out to USB

ISOLATION BARRIER down the middle of this block.

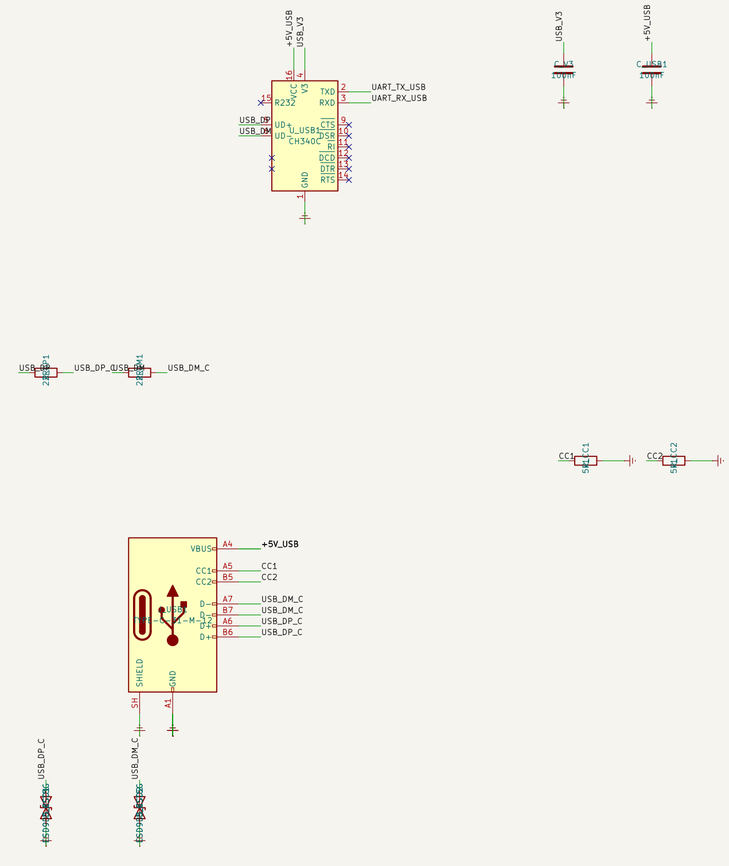
LEFT = bench-referenced (GND). RIGHT = laptop side (GNDREF).

No isolated DC-DC: the laptop's USB supplies the isolated side (D-056).




6N137 is fast enough for 115200 baud. The TLP521 / PC817 already in stock are NOT — do not substitute them without measuring first (O-6).

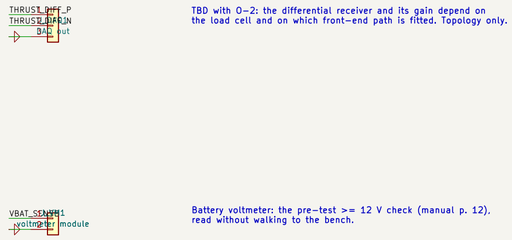
## 5. USB-C — laptop side, GNDREF domain




CC pull-downs 5k1 mark this as a UFP sink.
Data series 22R, one ESD diode per line.

## 6. MEASUREMENT OUTPUT — works with no software at all

Path 2: a voltage on a wire. Survives any software or USB failure.




TBD with O-2: the differential receiver and its gain depend on the load cell and on which front-end path is fitted. Topology only.

Battery voltmeter: the pre-test >= 12 V check (manual p. 12), read without walking to the bench.

Review owner: Emilio Guadarrama. Generated by fabrication/generate_schematics.py
TWO GROUND DOMAINS: GND = bench-referenced, GNDREF = laptop/USB. Never joined.
PRELIMINARY — NOT APPROVED. ARM, FIRE and E-STOP are mechanical, panel-mounted.

**Ignitia Rocket Lab**

Sheet: /
File: talavera-station.kicad_sch

***Title: TALAVERA v1 — Control Station Board***

Size: A2 | Date: 2026-11-06 | **Rev: A**
KiCad E.D.A. 10.0.5 | Id: 1/1

TBD values await O-2 (load cell) and O-6 (isolator part). ARM / FIRE / E-STOP carry no firmware.